

hatcher 3

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1 Kunneth Product

Define $\times : H^*(X; R) \times H^*(Y; R) \rightarrow H^*(X \times Y; R)$ by $a \times b = p^*(a) \cup p^*(b)$. Cup product is bilinear, so we have the induced map $\mu : H^*(X; R) \otimes H^*(Y; R) \rightarrow H^*(X \times Y; R)$. Let $H^*(X; R) \otimes H^*(Y; R)$ have the ring structure $(a \otimes b)(c \otimes d) = (-1)^{|b||c|} ac \otimes bd$. We check:

$$\begin{aligned} [(a_1 \otimes b_1)(a_2 \otimes b_2)](a_3 \otimes b_3) &= (-1)^{|b_1||a_2|} (a_1 a_2 \otimes b_1 b_2)(a_3 \otimes b_3) \\ &\quad (-1)^{|b_1||a_2| + |b_1 b_2||a_3|} a_1 a_2 a_3 \otimes b_1 b_2 b_3 \\ (a_1 \otimes b_1)[(a_2 \otimes b_2)(a_3 \otimes b_3)] &= (-1)^{|b_2||a_3|} (a_1 \otimes b_1)(a_2 a_3 \otimes b_2 b_3) \\ &= (-1)^{|b_2||a_3| + |b_1||a_2 a_3|} a_1 a_2 a_3 \otimes b_1 b_2 b_3 \end{aligned}$$

so associativity holds.

Now:

$$\begin{aligned} \mu((a \otimes b)(c \otimes d)) &= (-1)^{|b||c|} \mu(ac \otimes bd) = (-1)^{|b||c|} p_1^*(a \cup c) \cup p_2^*(b \cup d) \\ &= (-1)^{|b||c|} p_1^*(a) \cup p_1^*(c) \cup p_2^*(b) \cup p_2^*(d) \\ &= p_1^*(a) \cup p_2^*(b) \cup p_1^*(c) \cup p_2^*(d) \\ &= (a \times b)(c \times d) = \mu(a \otimes b) \mu(c \otimes d) \end{aligned}$$